Introduction to Programming

Lecture 5

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In the Previous Lecture

- Basic structure of C program
- Variables and Data types
- Operators
- ‘cout’ and ‘cin’ for output and input
- Braces

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Decision

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If Statement

If condition is true

statements

If Ali's height is greater then 6 feet

Then

Ali can become a member of the Basket Ball team

If Statement in C

```
If (condition)
statement ;
```

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If Statement in C

```
If ( condition )
{
   statement1 ;
   statement2 ;
    ⋮
}
```

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If statement in C

```
if (age1 > age2)
  cout<<"Student 1 is older than
   student 2" ;
```

Relational Operators

< less than

<= less than or equal to

== equal to

>= greater than or equal to

> greater than

!= not equal to

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Relational Operators

```
a != b;

X = 0;
X == 0;
```

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Example

```
#include <iostream.h>
main ( )
{
        int AmirAge, AmaraAge;
        AmirAge = 0;
        AmaraAge = 0;

        cout<<"Please enter Amir's age";
        cin >> AmirAge;
        cout<<"Please enter Amara's age";
        cin >> AmaraAge;

        if AmirAge > AmaraAge)
        {
         cout << "\n"<< "Amir's age is greater then Amara's age" ;
        }
}
```

Flow Chart Symbols

Start or stop

Process

Flow line

Continuation mark

Decision

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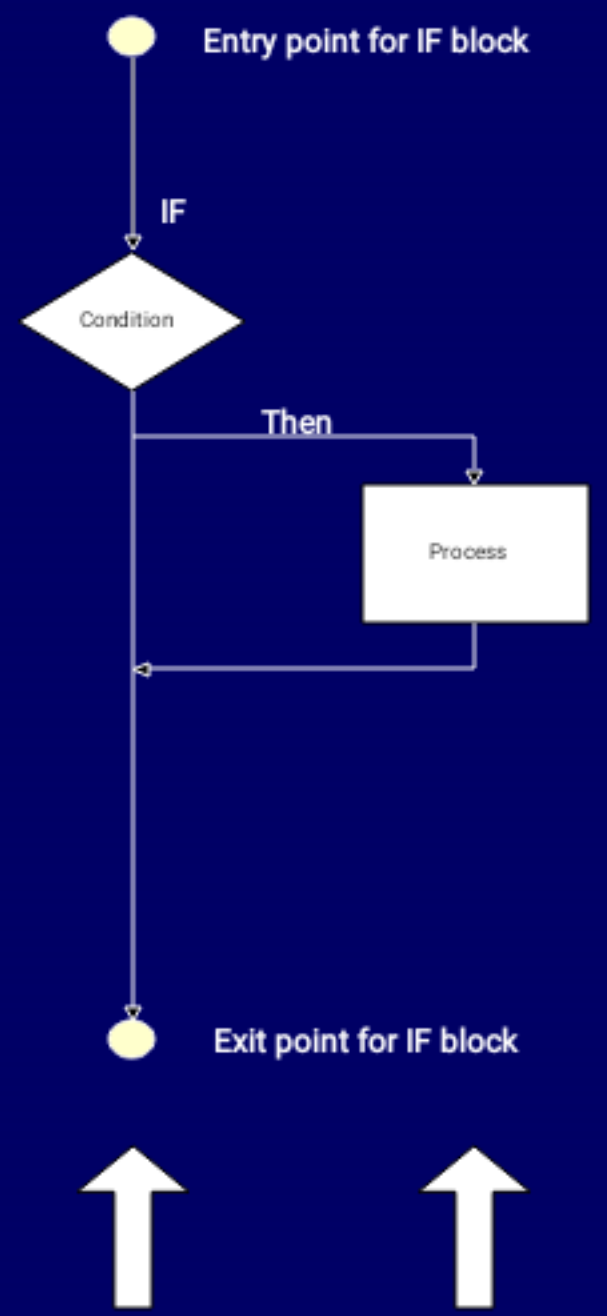
Flow Chart for if statement



Example

If the student age is greater than 18 or his height is greater than five feet then put him on the foot balll team

Else

Put him on the chess team

Logical Operators

AND &&

OR ||

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Logical Operators

If a is greater than b
AND c is greater than d

In C

```
if(a > b && c> d)
if(age > 18 || height > 5)
```

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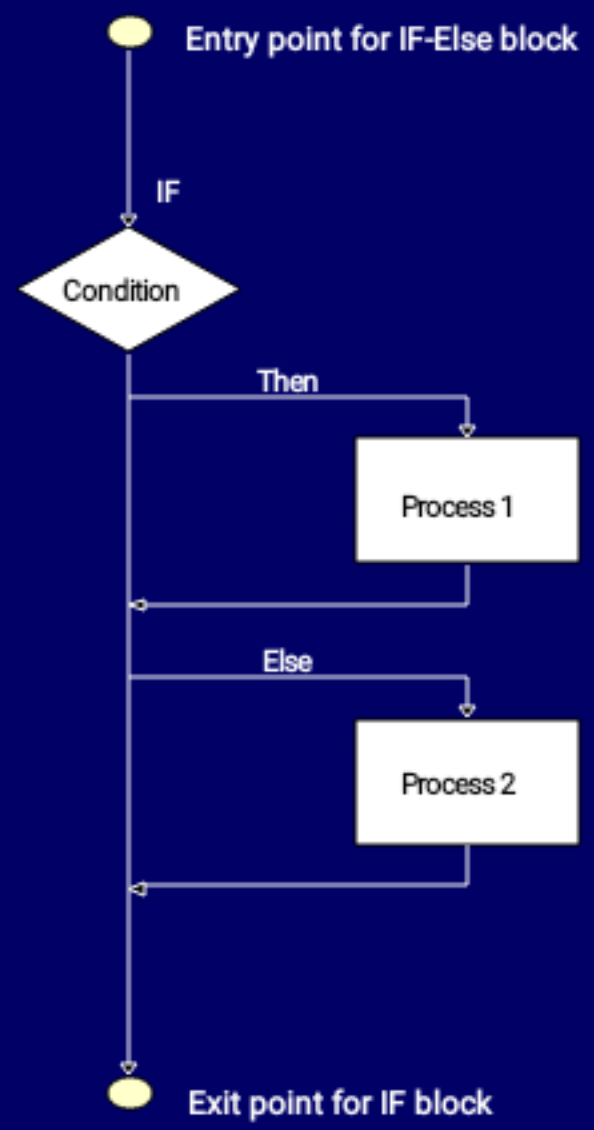
if-else

```
if (condition)
{
 statement ;
  -
  -
}
else
{
 statement ;
  -
  -
}
```

if-else



Note indentation from left to right

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Example

Code

```
if (AmirAge > AmaraAge)
{
    cout<< “Amir is older than Amara” ;
}

if (AmirAge < AmaraAge)
{
    cout<< “Amir is younger than Amara” ;
}
```

Example

Code

```
if AmirAge >  AmaraAge)
{
    cout<< “Amir is older than Amara” ;
}
else
{
    cout<<“Amir is younger than Amara” ;
}
```

Make a small flow chart of this program and see the one to one correspondence of the chart and the code

Example

Code

```
if AmirAge  >  AmaraAge)
{
    cout<< “Amir is older than Amara” ;
}
else
{
    cout<<“Amir is younger than or of the same ag
    as Amara” ;
}
```

Example

```
If (AmirAge != AmaraAge)
  cout << "Amir and Amara's Ages
  are not equal";
```

Unary Not operator !

- !true = false
- !false = true

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If (!(AmirAge > AmaraAge))

?

Example

```
if ((interMarks > 45) && (testMarks >= passMarks))
{
  cout << " Welcome to GC University";
}
```

Example

```
If(!((interMarks > 45) && (testMarks >= passMarks)))
```

?

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Nested if

```
If (age > 18)
{
    If(height  > 5)
    {
      :
    }
}
```

Make a flowchart of this nested if structure…

In Today's Lecture

- Decision
  - If
  - Else
- Flowcharts
- Nested if